UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	:	INTERACTIVE MULTIMEDIA
COURSE CODE	:	CSC413
EXAMINATION	:	NOV 2026 (MOCK)
TIME	:	3 HOURS

MARKING SCHEME NOTES

1. This marking scheme accompanies the CSC413 mock paper (NOV 2026). Answers show the expected marking points; all questions are deterministic and paper-derived; no lecture slides are used.

2. Marking points show the expected answer and the allocation of marks; unresolved slots would remain visible.

3. Part A = 15 x 1 mark; Part B = 15 x 1 mark; Part C = 30 x 1 mark; Part D = 4 x 10 marks; total = 100.

4. Questions: 64. Slots: 72. Placeholders: 0. No LLM call. Questions and answers are for local study use only.

5. Structure derived from 15 unique CSC413 past papers; no past-paper text is copied.

FOR MARKERS' USE ONLY

This marking scheme consists of 6 printed pages

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PART A - ANSWER KEY

Q	Ans	Statement	Explanation
A1	TRUE	A web browser displays hypertext documents.	Browsers render web pages.
A2	TRUE	Authoring tools help developers organise and edit multimedia elements.	Authoring tools assemble media.
A3	TRUE	A scanner is an input device.	A scanner captures documents into the computer.
A4	TRUE	Hypermedia combines multimedia with hyperlinks.	Hypermedia adds linked navigation to multimedia.
A5	FALSE	A vector image becomes pixelated when enlarged.	Vector images scale without pixelation.
A6	TRUE	A storyboard is part of project planning.	Storyboards plan the sequence.
A7	TRUE	Interactivity gives users control over content and flow.	Interactivity is a defining feature of many multimedia products.
A8	FALSE	3D animation does not use modelling.	3D animation relies on models.
A9	FALSE	MIDI files store digitized audio samples.	MIDI stores musical instructions, not samples.
A10	TRUE	A storyboard is a visual plan of a multimedia sequence.	Storyboards show the planned flow.
A11	TRUE	USB is a plug-and-play connection standard.	USB devices are recognised automatically.
A12	FALSE	Interactivity level is irrelevant when choosing an authoring tool.	The required interactivity affects tool choice.
A13	FALSE	A plotter is an input device.	A plotter produces output.
A14	TRUE	A prototype is an early model of a product.	Prototypes test ideas before production.
A15	FALSE	A multimedia project always uses every media element.	A project selects the elements it needs.

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PART B - ANSWER KEY

Q	Ans	Term	Definition
B1	H	Multimedia	integration of text, graphics, audio, video and animation
B2	C	Keyframe	important frame in an animation
B3	E	Sampling	converting sound into digital values
B4	D	Tweening	generating in-between frames
B5	I	Linear multimedia	presentation without user control of sequence
B6	M	Cel animation	hand-drawn frame-by-frame animation
B7	G	Hypermedia	linked multimedia elements
B8	K	Path animation	moving an object along a route
B9	N	MIDI	standard for musical notes and instructions
B10	B	Virtual reality	simulated environment
B11	F	Frame rate	frames displayed per second
B12	J	Chroma key	replacing a colour background
B13	A	Bitmap	image made of pixels
B14	O	Browser	software that displays web pages
B15	L	Vector	image made of mathematical objects

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PART C - ANSWER KEY

Q	Ans	Correct answer
C1	B	Smooth jagged edges
C2	C	Compiler
C3	C	Draws alternate lines of a frame
C4	A	Number of pixels in an image
C5	A	A video compression family
C6	C	Messages with a beginning and end
C7	B	WAV
C8	A	Permission to use a work
C9	A	Frequency
C10	B	Card-based
C11	A	Drawing input
C12	B	A visual sequence plan
C13	B	Play while downloading
C14	D	Loudness
C15	D	Text, graphics, audio, video and animation
C16	D	Another resource
C17	C	Screen display
C18	D	Automatic device recognition
C19	C	Printed output
C20	D	Drawing each frame by hand
C21	D	Scanner
C22	B	Frames per second
C23	B	Pixels
C24	C	Display web pages
C25	D	Along a predefined route
C26	D	Space between individual character pairs
C27	D	Merging of media and computing
C28	B	It combines visuals and sound
C29	C	Small strokes at character ends
C30	A	Links to other content

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PART D - WORKED ANSWERS

QUESTION 1

Topic: graphics

A product catalogue must show sharp images on screen and in print.

a) State TWO differences between bitmap and vector images.

Bitmap images use pixels and can pixelate; vector images use mathematical objects and scale cleanly.

(2 marks)

b) Describe TWO factors that affect image quality.

Resolution and colour depth both affect how much detail an image can show.

(4 marks)

c) Explain how colour model and resolution affect screen and print output.

Screens use RGB and print uses CMYK; print normally needs a higher resolution for sharp output.

(4 marks)

QUESTION 2

Topic: web hypermedia

A museum wants a website that presents its collection and lets visitors explore related exhibits.

a) State TWO differences between linear and non-linear navigation.

Linear follows a fixed sequence; non-linear lets users choose their own path.

(2 marks)

b) Describe TWO roles of hyperlinks in the website.

They connect related exhibits and let users move between pages.

(3 marks)

c) Explain how streaming and compression affect multimedia delivery on the web.

Compression reduces file size for faster transfer; streaming lets audio or video play while it downloads.

(5 marks)

QUESTION 3

Topic: animation

You are required to animate a company logo. The logo depicts a planet orbiting the sun.

a) State TWO animation file formats suitable for this project.

GIF and MP4 are suitable; SWF is also acceptable.

(3 marks)

b) Describe the steps of cel animation and ONE advantage of using it.

Draw each frame, capture the frames in sequence, then play them back; it gives direct artistic control.

(3 marks)

c) Differentiate 2-D and 2.5-D animation with ONE example for each.

2-D uses flat drawings such as a cartoon; 2.5-D adds depth to flat artwork, such as parallax layers in a game.

(4 marks)

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QUESTION 4

Topic: authoring interactivity

A company wants an interactive product demonstration that runs on a kiosk and on the web.

a) State TWO features to consider when choosing authoring software.

Interactivity support and export formats are important features.

(3 marks)

b) Explain the interactivity level suitable for the demonstration.

A medium level with menus and buttons lets users explore without overwhelming them.

(3 marks)

c) Explain TWO authoring tool categories appropriate for this project.

Card-based tools suit structured screens; timeline-based tools suit animation and media sequencing.

(4 marks)

END OF MARKING SCHEME

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